

Chapter 5

1

$$\alpha = (12)(45)$$

$$\alpha' = (12)(45)$$

$$\alpha^{-1} = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 3 & 5 & 4 & 6 \end{bmatrix}$$

b

$$\beta\alpha = (16532)(12)(45)$$

$$= (1)(26543)$$

$$= \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 6 & 2 & 3 & 4 & 5 \end{bmatrix}$$

$$\beta = (12)(45)(16532)$$

$$= (16453)$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 2 & 1 & 5 & 3 & 4 \end{bmatrix}$$

2

$$L_4 \alpha = (12345)(678), \quad \beta = (23847)(56)$$

a

$$\alpha = (12345)(678), \quad \beta = (23847)(56)$$

b

$$\alpha = (15)(14)(13)(12)(68)(67)$$

$$\beta = (24)(27)(26)(25)(56)$$

3

$$(1235)(413) = (15)(234)$$

b

$$(13256)(23)(46512) = (124)(35)$$

c

$$(12)(13)(23)(14) = (1423)$$

4

$$|(14)| = 2$$

b

$$|(147)| = 3$$

c

$$|(14762)| = 5$$

d

$$|(a_1 a_2 \dots a_n)| = n$$

5

$$|(124)(357)| = 3$$

b

$$|(124)(3567)| = 12$$

c

$$|(124)(35)| = 6$$

d

$$|(124)(2674)(9)| = 6$$

c. $| (1235)(24567) | = | (124)(3567) | = 12$

f. $| (345)(245) | = | (25)(34) | = 2$

6. $| (12)(356) | = 6$

b. $| (1753)(264) | = 12$

7. 12

8. $(135) = (15)(13) \Rightarrow$ even permutation

b. $(1356) = (16)(15)(13) \Rightarrow$ odd permutation

c. $(13567) = (17)(16)(15)(13) \Rightarrow$ even permutations

d. $(12)(134)(152) = (12)(14)(13)(12)(15) \Rightarrow$ odd permutation

e. $(1243)(3521) = (13)(14)(12)(31)(32)(35) \Rightarrow$ even permutation

9. (6)

$(5)(1)$

$(4)(2)$

$(4)(1)(1)$

$(3)(3)$

$(3)(2)(1)$

$(2)(2)(2)$

$(2)(2)(1)(1)$

$(2)(1)(1)(1)(1)$

$(1)(1)(1)(1)(1)(1)$

Orders of element in $S_6 = 6, 5, 4, 3, 2, 1$

b. In A_6

$(1)(1)(1)(1)(1)(1) = 0$ transposition is even

$(3)(1)(1)(1) = 2$ transpositions is even

$(5)(1) = 4$ transposition is even

$(3)(3) = 4$ transposition is even

$(4)(2) = 4$ transposition is even

Orders of element in $A_6 = 1, 3, 4, 5$

In A_7

$(1)(1)(1)(1)(1)(1)(1) = 0$ transposition is even

$(3)(1)(1)(1)(1) = 2$ transposition is even

$(5)(1)(1) = 4$ transposition is even

$(7) = 6$ transposition is even

$(3)(3)(1) = 4$ transposition is even

$(4)(2)(1) = 4$ transposition is even

Order of element in $A_7 = 1, 3, 4, 5, 7$

10 Consider disjoint cycle of $(3)(5) = \alpha$

This contains 6 transpositions

$\therefore \alpha \in A_8$ with order 15

11 Consider disjoint cycle of $(3)(5)(2)(2) = \beta$

This decomposed contains $2+4+1+1 = 8$ even transposition

$\therefore \beta \in A_{12}$ where $|\beta| = 30$

15 For any single n -cycle, there is $n-1$ transpositions

If n is odd, there exist even permutation

If n is even, there exist odd permutation

16 Let α be an even permutation

This means $\alpha = T_1 T_2 \dots T_{2k}$ for some $k \in \mathbb{Z}$

where T_i is a transposition

We know $(ab)^{-1} = b^{-1}a^{-1}$

Applying same principle

$$\alpha^{-1} = T_{2k}^{-1} \cdot T_{2k-1}^{-1} \dots T_1^{-1}$$

We also know that when decomposing placement of these single cycle don't matter because $(12) = (21)$

so each transposition is its own inverse

$$T_i^{-1} = T_i \text{ for all } i$$

thus $\alpha^{-1} = T_{2k} T_{2k-1} \dots T_1$

α^{-1} consists of $2k$ transposition which is even

$\therefore$ Since we expressed α^{-1} as a product of an even number of transpositions, α^{-1} is an even permutation

b. Let α be some odd permutation

This means α can be expressed as product of $2k+1$ transpositions for some $k \in \mathbb{Z}$.

$$\alpha = T_1 T_2 \dots T_{2k+1} \text{ where } T_i \text{ is a transposition}$$

Recall $(ab)^{-1} = b^{-1} a^{-1}$

$$\alpha^{-1} = (T_1 T_2 \dots T_{2k+1})^{-1} = T_{2k+1}^{-1} T_{2k}^{-1} \dots T_1^{-1}$$

Since each transposition is its own inverse because

$$(T_i)^{-1} = T_i, \quad \forall i$$

Thus $\alpha^{-1} = T_{2k+1} T_{2k} \dots T_1$ consists of $2k+1$ transpositions which is odd

$\therefore \alpha^{-1}$ is an odd permutation

17. Let $G = S_n$ be symmetric group of permutation on n and H be set of even permutations

Closure: Let $\alpha, \beta \in H$. This means $\alpha = T_1 T_2 \dots T_{2k}$

and $\beta = \gamma_1 \gamma_2 \dots \gamma_{2l}$ for some $k, l \in \mathbb{Z}$, where T_i, γ_i transpos.

Product $\alpha\beta = (T_1 T_2 \dots T_{2k})(\gamma_1 \gamma_2 \dots \gamma_{2l})$ which contains $2k+2l$ transpositions where $x = k+l$ with $2x$ being even.

$\therefore \alpha\beta \in H$ and H is closed

$\therefore H$ is a subgroup of S_n by finite subgroup test

19. $\alpha\beta$ is even $\Rightarrow \alpha, \beta$ are both even or both odd

If α, β are both even or odd $\Rightarrow \alpha\beta$ is even

Proof:

1. Let $\alpha\beta$ be even and k equal number of decomposed cycles needed to express α while m for β respectively.

We know $\alpha\beta = (\text{transpositions for } \alpha)(\text{transposition for } \beta)$
 Thus $\alpha\beta$ consists of $k+m$ transpositions. Since ab is even, we know $k+m$ must be even also

Case 1: When k is even, we know α is an even permutation and m must be even also for $k+m$ to be even. Thus β is an even permutation also

Case 2: When k is odd, we know α is an odd permutation and m must be odd also for $k+m$ to be even. Thus β is also an odd permutation

$\therefore$ If $\alpha\beta$ is even, then both α and β must be either both even or both odd

ii. Let α & β be both even or odd.

Both even: If α and β can be expressed as an even number of transpositions then $\alpha\beta$ consists of even + even number of transpositions
 Thus $\alpha\beta$ is an even permutation

Both odd: If α and β can be expressed as an odd number of transpositions then $\alpha\beta$ consists of odd + odd number of transpositions.

Thus $\alpha\beta$ is an even permutation

$\therefore \alpha\beta$ is even iff both α & β are either both even or both odd

22 $(a_1 a_2 \dots a_n)^{-1} = (a_n a_{n-1}, \dots, a_2 a_1)$

25 Disprove two step subgroup test

Let H be subgroup of all permutations

identity: This fails because E has 0 transpositions which is even. Thus $E \notin H$.

closure: Let $a, b \in H$. This means α and β consist of

k and m transpositions respectively where k & m are odd.

Product ab can then be expressed as having $k+m$ transpositions.

Since both k & m are odd, we have $l = k+m$ is even. Thus ab is an even permutation and it is not closed. $\therefore$ Since the set of odd permutations in S_n violates 2 criteria of the 2st, it is not a subgroup of S_n .

31. Let $b^4 = (2143567)$ in S_7

$$|b^4| = 7$$

$$\text{We know } |b^4| = \frac{|b|}{\gcd(|b|, 4)}$$

$$|b| = |b^4| \gcd(|b|, 4)$$

Possible values for $\gcd(|b|, 4)$: 1, 2, 4

$$\therefore \text{If } \gcd(|b|, 4) = 1, |b| = 7 \cdot 1 = 7$$

$$\therefore \text{If } \gcd(|b|, 4) = 2, |b| = 7 \cdot 2 = 14$$

$$\therefore \text{If } \gcd(|b|, 4) = 4, |b| = 7 \cdot 4 = 28$$

$$\text{So } |b| \in \{7, 14, 28\}$$

Let $a = (b)^4$, find b such that $b = a^k$ $k \in \mathbb{Z}$

$$b^4 = (a^k)^4 = a^{4k} = a$$

$$a^{4k} = a$$

$$a^{4k} \div a = e$$

$$a^{4k-1} = e$$

$$\text{Since } |a| = |b^4| = 7$$

$$a^7 = e$$

$$\text{Thus } a^{4k-1} = (a^7)^s \text{ for some } s \in \mathbb{Z}$$

$$4k-1 \equiv 0 \pmod{7}$$

$$4k \equiv 1 \pmod{7}$$

Find k ;

$$K = 1 : 4 \times 1 = 4 \not\equiv 1 \pmod{7}$$

$$K = 2 : 4 \times 2 = 8 \equiv 1 \pmod{7}$$

So $K = 2 + 7x$ for some $x \in \mathbb{Z}$ in bounds of S_7

$\therefore K$ must equal 2

$$\therefore b = a^2$$

$$a = (2143567)$$

$$\begin{aligned} b &= (2143567)(2143567) \\ &= (2457136) \end{aligned}$$

32. Let $\beta = (123)(145)$. Find β^{99}

$$\beta = (14523)$$

$$\text{Thus } |\beta| = 5$$

$$\text{Now } 99 \pmod{5} = 4 \text{ and } \beta^{99} = \beta^4$$

$$\therefore \beta^4 = (13254)$$

33. Let $x^2 = (4)(2)$ in S_{10}

$$|x^2| = 4$$

$$\therefore |x| = 8$$

So we have x to be an 8 cycle in S_{10}

$$\text{Denote } x = (b_1 b_2 b_3 b_4 b_5 b_6 b_7 b_8)$$

$$x^2 = (b_1 b_3 b_5 b_7)(b_2 b_4 b_6 b_8) \neq (4)(2)$$

36. Let $\beta = (1, 3, 5, 7, 9, 8, 6)(2, 4, 10)$. [Find $b^n = b^5$]

$$|\beta| = \text{LCM}(7, 3) = 21$$

$$\text{Thus } b^{-5} = b^{21-5} = b^{16}$$

$$\text{So } b^n = b^{16}$$

$$b^{n-16} = \beta$$

$$\text{Thus } |b| \mid n-16$$

$$n-16 \equiv 0 \pmod{21}$$

$$n \equiv 16 \pmod{21}$$

To find the smallest n in this congruence class

$$n = 16 + 21p \quad \text{for some } p \in \mathbb{Z}$$

$$\text{At } p = 0$$

$$n = 16 + 21(0) = 16$$

Smallest n for $b^n = b^{-1}$ is 16

37. Let $\alpha = (1, 3, 5, 7, 9)(2, 4, 6)(8, 10)$

The cycle of length k , $(\text{cycle})^m$ will equal single cycle also if $\gcd(k, m) = 1$ or split into $\gcd(k, m)$ cycles of length $\frac{k}{\gcd(k, m)}$

We know $|\alpha| = 30$ and

$$\alpha^m = (13579)^m \cdot (246)^m \cdot (8, 10)^m$$

Since each cycle are disjoint each cycle to powers will behave independently.

Looking at the conditions, and separate cycles

- i. $(13579)^m$ remains a single 5 cycle if $\gcd(m, 5) = 1$
- ii. (246) is a cycle of length 3 not 5 so $(246)^m$ becomes ϵ if $3|m$; $m \equiv 0 \pmod{3}$
- iii. $(8, 10)$ is a 2-cycle so $(8, 10)^m = \epsilon$ if $2|m$; $m \equiv 0 \pmod{2}$

Finding m :

- m must be a multiple of $\text{LCM}(2, 3) = 6$, so $m \equiv 0 \pmod{6}$
- from ci) $m \nmid 5$ are coprime so $m \not\equiv 0 \pmod{5}$

Thus m is any positive integer where:

$$m = 6k \quad \text{and} \quad \gcd(m, 5) = 1 \quad \text{so that} \quad (246)^m \neq \epsilon \quad (8, 10)^m = \epsilon$$

$$\therefore m = \{6, 12, 18, 24, \dots\}$$

38. Let $\mathcal{H}$ be set of permutations in S_3 that fixes 1 & 2 nonempty. Since $\mathcal{E}(1) = 1$ and $\mathcal{E}(3) = 3$, $\mathcal{E} \in \mathcal{H}$

closure: Let $\alpha, \beta \in \mathcal{H}$ and $\gamma = \alpha\beta$.

$$\gamma(1) = \alpha\beta(1) = \alpha[1] = 1$$

$$\gamma(3) = \alpha\beta(3) = \alpha[3] = 3$$

Thus $\alpha\beta$ is in $\mathcal{H}$ and $\mathcal{H}$ is closed under composition.

- $\therefore H$ is a subgroup of S_5 by finite subgroup test
- b. Since $H \leq S_5$ that fixes 1, 3, H is all possible permutations on set $\{2, 4, 5\}$
 $|H| = 3! = 6$
- c. Yes for any S_n ; if H fixes 2 elements
 $|H| = (n-2)!$
- d. Let $H = \{\pi \in A_n \mid \pi(1) = 1 \text{ \& } \pi(3) = 3\}$
 This means H consists of even permutations on remaining $n-2$ elements
 $|H| = \frac{(n-2)!}{2}$

45. Let $H = \{\pi \in S_n \mid \pi(1) = 1 \text{ or } 2 \text{ and } \pi(2) = 1 \text{ or } 2\}$
non empty: $e(1) = 1, e(2) = 2, e \in H$
closure: Let $\alpha, \beta \in H$ and $\gamma = \alpha \cdot \beta$
 $\gamma(1) = \alpha \cdot \beta(1) = \alpha(\{1, 2\})$ which is a closed mapping
 $\gamma(2) = \alpha \cdot \beta(2) = \alpha(\{1, 2\})$ which is also closed
 Thus $\alpha\beta \in H$ and H is closed under function composition
 $\therefore H$ is a subgroup by finite subgroup test

- b. In H , we know each permutation π satisfies
 $\pi(1): \pi(1) \in \{1, 2\}$
 $\pi(2): \pi(2) \in \{1, 2\}$
 We know function composition is one-to-one and onto, so $\pi(1) \neq \pi(2)$
 so, $\pi(1)$ and $\pi(2)$ are 1 and 2 in any sequence
 Now, for permutations on remaining elements:
 $\Rightarrow n-2$ elements; we determined # of ways to assign $\pi(1)$ and $\pi(2) = 2$
 $|H| = (n-2)! \times 2$